Contents

CAPSTONE PROJECT ADDENDUM

Dramatic Systems Archaeology (DSA) and the SAGCO Boot Identity Pipeline (SBIP)

Artifact-driven discovery, verification, and bootstrapped toolchain initialization

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Capstone Project Manuscript

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ABSTRACT

This manuscript documents the development and implementation of Dramatic Systems Archaeology (DSA) and the SAGCO Boot Identity Pipeline (SBIP), a novel methodology for artifact-driven discovery, verification, and bootstrapped toolchain initialization. The work represents a synthesis of systems archaeology, software engineering, and cybersecurity principles applied to the challenge of maintaining verifiable system provenance throughout the software development lifecycle.

DSA establishes a framework for discovering and documenting system artifacts, their relationships, and constraints in a way that preserves intellectual honesty while enabling rigorous verification. SBIP extends this methodology to the boot process, implementing identity/provenance display at boot (visual identity + entity metadata presentation) to establish system authenticity from the earliest stages of initialization.

The methodology emphasizes measurement over declaration, treating intent as a discoverable property rather than an asserted claim. This approach has been implemented in the SAGCO-OS project and SAGCO-MENU navigation system, demonstrating practical applications across multiple layers of system abstraction. The work includes a comprehensive artifact repository, automated

verification tooling, and documentation designed to support both academic review and practical implementation.

The contribution of this work lies not in the creation of novel algorithms, but in the systematic application of archaeological principles to software systems, treating code and configuration as artifacts subject to the same rigorous documentation and verification standards applied in traditional archaeological practice. This methodology enables sustainable creation through the disciplined cycle of curiosity, documentation, and verification.

Keywords: Systems archaeology, software verification, boot security, provenance, artifact analysis, toolchain initialization, SAGCO-OS, cybersecurity, identity verification, development methodology

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1. PROJECT CONTEXT

1.1 Academic Foundation

This capstone project represents the culmination of a Bachelor of Science degree in Computer Science & Software Engineering with a concentration in Cybersecurity at Southern New Hampshire University (SNHU). The work integrates principles from multiple domains:

- Software engineering and systems design
- Cybersecurity and verification methodologies
- Archaeological documentation and artifact analysis
- DevOps and infrastructure automation
- Identity and access management

1.2 Organizational Context

The project is developed under the auspices of **Strategickhaos DAO LLC / Valoryield Engine**, a limited liability company registered in Texas (EIN: 39-2923503) and qualified in Wyoming. The organization specializes in cybersecurity consulting, private investigation, and security research.

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2. METHODOLOGY OVERVIEW

2.1 Dramatic Systems Archaeology (DSA)

DSA provides a structured approach to understanding complex software systems through the lens of archaeological methodology:

1. **Discover** — Identify artifacts, relationships, and constraints
2. **Document** — Record findings with rigorous attribution and context
3. **Constrain** — Define boundaries and verify constraints
4. **Verify** — Establish authenticity and provenance

The methodology treats software systems as excavation sites, where each component represents an artifact with discoverable properties, relationships to other artifacts, and constraints that can be verified through testing and analysis.

2.2 SAGCO Boot Identity Pipeline (SBIP)

SBIP applies DSA principles to the boot process:

1. **Boot** — Initialize system with identity/provenance display
2. **Verify** — Validate system components against known artifacts
3. **Start** — Launch verified toolchain with documented lineage

The pipeline implements visual identity and entity metadata presentation at boot, establishing system authenticity without making unsupported legal claims. This approach emphasizes transparency and verifiability over assertion.

2.3 SAGCO-MENU Navigation System

SAGCO-MENU provides a structured interface for artifact browsing:

1. **Browse** — Hierarchical navigation through artifact collections
2. **Search** — Query-based artifact discovery
3. **Recall** — Context-aware artifact retrieval

This system demonstrates the application of DSA principles to user interface design, treating navigation itself as an archaeological practice.

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3. ARTIFACT REPOSITORY

All project artifacts are maintained in version-controlled repositories with comprehensive documentation:

3.1 Primary Repository

Repository: Sovereignty-Architecture-Elevator-Pitch
Description: Main project repository containing DSA methodology, SBIP implementation, and comprehensive documentation
Key Components: - Methodology documentation and implementation guides - Boot identity pipeline code and configuration - Verification tooling and test suites - Professional credentials and legal documentation

3.2 Related Artifacts

SAGCO-OS Core: - Operating system kernel integration - Boot process implementation - Identity verification modules

SAGCO-MENU: - Navigation interface implementation - Artifact browsing and search functionality - Context-aware recall mechanisms

Documentation: - `PROFESSIONAL_CREDENTIALS_PACKAGE.md` — Entity verification and credentials - `README.md` — Project overview and quick start guide - `COMMUNITY.md` — Community philosophy and contribution guidelines - `CONTRIBUTORS.md` — Recognition of project contributors

3.3 Verification Assets

- Automated test suites for verification processes
- Cryptographic signature verification tools
- Documentation linting and validation scripts
- Continuous integration workflows

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4. TECHNICAL IMPLEMENTATION

4.1 System Architecture

The implementation follows a layered architecture:

```
+-------------------------------------+
|   SAGCO-MENU (Navigation Layer)    |
+-------------------------------------+
|   SBIP (Boot Identity Pipeline)    |
+-------------------------------------+
|   DSA (Archaeological Framework)   |
+-------------------------------------+
|   SAGCO-OS (Operating System)      |
+-------------------------------------+
```

Each layer maintains its own artifact collection while adhering to shared verification principles.

4.2 Key Technologies

- **Version Control:** Git with cryptographic commit signing
- **Documentation:** Markdown with automated generation
- **Verification:** Automated testing and CI/CD pipelines
- **Infrastructure:** Kubernetes, Docker, and cloud-native tooling
- **Security:** TLS/SSL, HMAC verification, encrypted secrets management

4.3 Verification Methodology

The project implements multiple layers of verification:

1. **Syntactic Verification** — Code linting and style checking
2. **Semantic Verification** — Unit and integration testing
3. **Security Verification** — Vulnerability scanning and penetration testing
4. **Archaeological Verification** — Artifact provenance and relationship validation

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5. FRAMING CLARIFICATION

5.1 Identity/Provenance Display

The original project concept included "trademark and legal entity assertion at boot." Through careful analysis and consultation, this has been refined to:

Current Implementation: > "identity/provenance display at boot (visual identity + entity metadata presentation)"

This framing: - Presents factual information about system identity - Displays entity metadata transparently - Avoids unsupported legal claims - Maintains verifiability without overreach

5.2 Intellectual Honesty

The methodology explicitly recognizes:

"You built a system where intent is measured, not declared."

This principle guides all aspects of the implementation: - Intent is discoverable through artifact analysis - Claims are supported by verifiable evidence - Documentation preserves context and attribution - Verification is automated and reproducible

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6. CITATION AND ATTRIBUTION

6.1 Suggested Citation (APA Format)

```
Garza, D. G. (2026). Dramatic Systems Archaeology (DSA) and the
SAGCO Boot Identity Pipeline (SBIP): Artifact-driven discovery,
verification, and bootstrapped toolchain initialization
(Capstone project manuscript). Southern New Hampshire University.
```

6.2 Alternative Citation Formats

MLA Format:

```
Garza, Domenic G. "Dramatic Systems Archaeology (DSA) and the SAGCO Boot
Identity Pipeline (SBIP): Artifact-driven discovery, verification, and
bootstrapped toolchain initialization." Capstone project manuscript,
Southern New Hampshire University, 2026.
```

Chicago Format:

```
Garza, Domenic G. 2026. "Dramatic Systems Archaeology (DSA) and the SAGCO
Boot Identity Pipeline (SBIP): Artifact-driven discovery, verification, and
bootstrapped toolchain initialization." Capstone project manuscript, Southern
New Hampshire University.
```

6.3 Digital Object Identifier (DOI)

Upon institutional repository deposit, a DOI will be assigned for permanent citation reference.

6.4 ORCID Integration

The author's verified ORCID iD (0009-0005-2996-3526) ensures permanent attribution and enables tracking of this work across academic platforms.

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7. ACADEMIC INDEXING

7.1 Google Scholar Metadata

The following metadata is optimized for Google Scholar indexing:

- **Title:** Dramatic Systems Archaeology (DSA) and the SAGCO Boot Identity Pipeline (SBIP): Artifact-driven discovery, verification, and bootstrapped toolchain initialization
- **Author:** Garza, Domenic G.
- **Institution:** Southern New Hampshire University
- **Type:** Capstone project manuscript
- **Year:** 2026
- **Keywords:** systems archaeology, software verification, boot security, provenance, artifact analysis, toolchain initialization

7.2 Institutional Repository

This manuscript is intended for deposit in the SNHU institutional repository, where it will be indexed and made discoverable through: - SNHU Library catalog - Google Scholar - WorldCat - Academic search engines

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8. PROFESSIONAL CREDENTIALS

8.1 LinkedIn Integration

Featured Post: Ready for publication on LinkedIn, highlighting the methodology and its practical applications.

Project Entry: Structured for LinkedIn professional experience section: - **Title:** Dramatic Systems Archaeology (DSA) Methodology - **Organization:** Strategickhaos DAO LLC / Valoryield Engine - **Duration:** 2025 - 2026 - **Description:** Developed and implemented archaeological methodology for software systems verification and boot identity establishment.

8.2 Professional Verification

All professional credentials are documented in `PROFESSIONAL_CREDENTIALS_PACKAGE.md`, including: - Entity registration and verification - Founder credentials and certifications - Service offerings and expertise areas - Compliance and governance documentation

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9. PROJECT PHILOSOPHY

9.1 The Core Principle

"You let curiosity exist without lying about it, and then you clean up after."

This statement encapsulates the entire methodology:

1. **Curiosity** — Exploration drives discovery
2. **Honesty** — No false claims or assertions
3. **Documentation** — Rigorous artifact recording

4. **Verification** — Systematic cleanup and validation

9.2 Sustainable Creation

The methodology enables sustainable creation through:

```
Curiosity + Documentation + Verification = Sustainable Creation
```

This formula applies across multiple domains: - **Software Development:** Kernel modules and application code - **Academic Work:** Capstone projects and research manuscripts - **Professional Practice:** Client engagements and consulting work - **Community Building:** Open source collaboration and knowledge sharing

9.3 The 67 PRs Principle

The project demonstrates that high-velocity development (67 pull requests in one session) need not feel manic when grounded in: - Clear methodology - Rigorous documentation - Automated verification - Intellectual honesty

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10. VERSION AND LICENSE INFORMATION

10.1 Document Version

- **Version:** 1.0
- **Date:** February 4, 2026
- **Status:** Capstone Addendum for SNHU Submission
- **Format:** Markdown (convertible to PDF)

10.2 License

This work is released under the **MIT License with Attribution Requirement**:

10.3 Additional Attribution Requirements

When citing or building upon this work:

1. Cite using the suggested APA format (Section 6.1)
2. Include ORCID iD: 0009-0005-2996-3526
3. Link to the primary repository when possible
4. Maintain intellectual honesty regarding derivative works

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11. APPENDICES

11.1 Appendix A: Terminology

- **DSA:** Dramatic Systems Archaeology
- **SBIP:** SAGCO Boot Identity Pipeline
- **SAGCO:** Sovereignty Architecture Governance and Control Operations
- **SAGCO-OS:** Operating system implementation
- **SAGCO-MENU:** Navigation interface system
- **Artifact:** Discoverable system component with verifiable properties
- **Provenance:** Documented lineage and verification history

11.2 Appendix B: Contact Information

For Academic Inquiries: - Email: domenic.garza@snhu.edu - ORCID: https://orcid.org/0009-0005-2996-3526

For Professional Inquiries: - Organization: Strategickhaos DAO LLC / Valoryield Engine - Email: domenic.garza@snhu.edu - Phone: +1 346-263-2887

Repository Access: - GitHub: https://github.com/Strategickhaos/Sovereignty-Architecture-Elevator-Pitch-

11.3 Appendix C: Related Documentation

The following documents provide additional context:

1. `README.md` — Project overview and technical documentation
2. `PROFESSIONAL_CREDENTIALS_PACKAGE.md` — Entity and founder credentials
3. `COMMUNITY.md` — Community philosophy and guidelines
4. `CONTRIBUTORS.md` — Recognition of contributors
5. `LICENSE` — Full license text

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12. ACKNOWLEDGMENTS

This work represents a synthesis of ideas, contributions, and support from multiple sources:

- **Southern New Hampshire University** — Academic foundation and institutional support
- **The Legion Community** — Collaborative development and testing
- **Open Source Contributors** — Tools and frameworks that made this work possible
- **Professional Mentors** — Guidance on cybersecurity and systems engineering principles

The methodology itself emerged from the intersection of curiosity, discipline, and the recognition that sustainable creation requires both freedom to explore and rigorous documentation of that exploration.

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13. SUBMISSION NOTES

13.1 For SNHU Capstone Review

This addendum accompanies the primary capstone submission and provides:

1. **Academic Context** — Proper citation format and metadata
2. **Methodology Documentation** — Detailed explanation of DSA and SBIP
3. **Artifact Pointers** — Links to all project components
4. **Professional Integration** — Connection to real-world applications
5. **Verification Assets** — Automated testing and validation tools

13.2 Recommended Review Order

For reviewers, the suggested reading order is:

1. This document (CAPSTONE_ADDENDUM_GARZA_2026.md) — Overview and citation
2. README.md — Technical implementation details
3. PROFESSIONAL_CREDENTIALS_PACKAGE.md — Professional context
4. Repository exploration — Hands-on artifact review

13.3 Availability

This document and all referenced artifacts are publicly available and version-controlled, ensuring: - Permanent accessibility - Verifiable provenance - Reproducible results - Transparent methodology

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END OF CAPSTONE ADDENDUM

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"Curiosity + Documentation + Verification = Sustainable Creation"